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JESUS MINJARES

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PROFESSIONAL SUMMARY

Embedded Software Engineer with active DOE Q Clearance and 3+ years at Sandia National Laboratories developing real-time embedded systems, HWIL infrastructure, and avionics interfaces for mission-critical applications. Skilled in low-level C/C++ development, FPGA-based sensor emulation, serial and I²C interface development, hardware bring-up, and system integration across lab and test environments. Currently pursuing an M.S. in Artificial Intelligence at UT Austin (4.0 GPA).

TECHNICAL EXPERIENCE

Sandia National Laboratories — Albuquerque, NM
Embedded Software Engineer — Nov. 2022 – Present

- Implemented an FPGA-based IMU emulator on Speedgoat using MATLAB/Simulink and HDL Coder, generating a 2400 Hz SDLC-framed bitstream for avionics integration testing.
- Engineered an isochronous serial IMU emulator using Microgate SyncLink, providing a 1 MHz bidirectional interface to the flight computer for chassis-based sensor emulation.
- Wrote a C++ I²C bit-banging driver for the NI 6509, enabling deterministic multi-device communication within the test chassis without third-party dependencies.
- Led development and sustainment of seven chassis-specific iHawk models, coordinating with ground software, flight software, and HWIL electrical teams to resolve signal, protocol, and timing discrepancies.
- Built validation software for 8 HWIL chassis and supported hardware verification through lab-based testing of voltage monitoring, relay control, and system interfaces, reducing integration risk.
- Hardened the OTA firmware update process by implementing dual-stage hardware version validation, preventing incompatible deployments and reducing remote bricking risk.
- Modernized a legacy firmware repository by migrating to GitLab CI/CD, enabling automated multi-target builds and improving build reproducibility.

Aerospace Center (cSETR) — El Paso, TX
Graduate Research Assistant — Aug. 2021 – Jun. 2022

- Co-led firmware development for a 3U CubeSat on a 5-member multidisciplinary team spanning mechanical, electrical, and software subsystems.
- Authored C firmware for 3 flight microcontrollers supporting data acquisition, control loops, and fault handling.
- Designed, built, and validated custom flight hardware PCBs, including soldering, bring-up, and debugging using oscilloscopes and DMMs.
- Introduced Git-based version control and Doxygen API documentation, standardizing development practices and